Incr., Decr., Local Extrema

Section 3.3

9/24/25

A: What we should know:

1) $f(x)$ is increasing when $f'(x) > 0$
2) $f(x)$ is decreasing when $f'(x) < 0$

LOCAL Extrema: 1^{st} Derivative Test

1) $f(x)$ has a <u>local max</u> when $f'(x)$ changes from positive to negative
2) $f(x)$ has a <u>local min</u> when $f'(x)$ changes from negative to positive.

Ex1:

@w incr./decr. local max/min

Step 1

$f(x) = x^3 - 4x$

$f'(x) = 3x^2 - 4 = 0$

$x = \pm\frac{2}{\sqrt{3}}$

$f'(x)$: number line with -5, 0, 5 marked; $+$ before $-\frac{2}{\sqrt{3}}$, $-$ between $-\frac{2}{\sqrt{3}}$ and $\frac{2}{\sqrt{3}}$, $+$ after $\frac{2}{\sqrt{3}}$

Step 2

Plug in between range, find our signs

Conclusion

$f(x)$ is increasing: $(-\infty, -\frac{2}{\sqrt{3}}) \cup (\frac{2}{\sqrt{3}}, \infty)$

because $f'(x) > 0$

$f(x)$ is decreasing: $(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$

because $f'(x) > 0$

$f(x)$ is local max: $\left(-\frac{2}{\sqrt{3}}, \left(-\frac{2}{\sqrt{3}}\right)^3 - 4\left(\frac{-2}{\sqrt{3}}\right)\right)$

because $f'(x)$ changes from Positive to Negative

Ex2: $x\sqrt{4-x}$

STEP ONE

$f'(x) = \sqrt{4-x} + \frac{1}{2}x(4-x)^{-1/2}(-1) = 0$

$\sqrt{4-x} - \frac{x}{2\sqrt{4-x}}$

$\frac{2(4-x) - x}{2\sqrt{4-x}} = \frac{8-3x}{2\sqrt{4-x}} = 0$

$x = 8/3$

$x = 4$

Step 2

f': number line with $8/3$ and 4 marked; $+$ before $8/3$, $-$ between $8/3$ and 4

f incr. : $(-\infty, 8/3)$ because $f'(x) > 0$

f decr: $(8/3, 4]$ because $f'(x) < 0$

f local max: @ $x = 8/3$, $f'(x)$ goes from $+$ to $-$

f local min: N/A

EX3: $f(x) = \frac{x^2}{2x-1}$

$f'(x) = \frac{2x(2x-1) - x^2(2)}{(2x-1)^2}$

$\frac{-2x + 4x^2 - 2x^2}{(2x-1)^2}$

$\frac{2x^2 - 2x}{(2x-1)^2} = 0$

$\frac{2x(x-1)}{(2x-1)^2} = 0$

$x = 0,\ x = 1,\ x = \frac{1}{2}$

$f'(x)$: number line with 0, $\frac{1}{2}$, 1 marked; $+$ before 0, $-$ between 0 and $\frac{1}{2}$, $-$ between $\frac{1}{2}$ and 1, $+$ after 1

$f(x)$ incr: $(-\infty, 0) \cup (1, \infty)$ b/c $f'(x) > 0$

$f(x)$ decr: $(0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$ b/c $f'(x) < 0$

$f(x)$ local max: $(0, 0)$ $f'(x)$ goes from $+$ to $-$

$f(x)$ local min: $(1, 1)$ $f'(x)$ goes from $-$ to $+$
